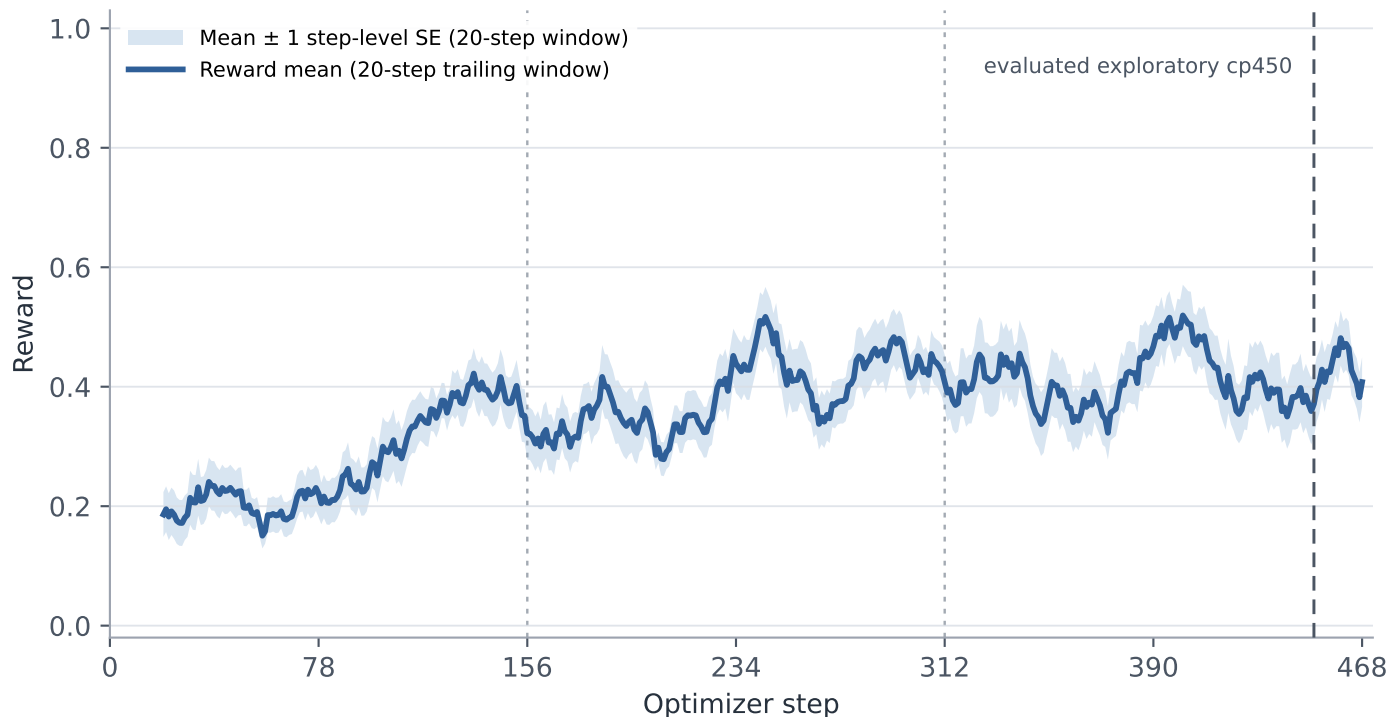


Model chk-3 GRPO: Reward Mean with ± 1 SE

Repository artifact chk4; 20 optimizer-step clusters per trailing window; 8 completions per step

Epoch reward mean $\pm$ step-level SE: E1 0.269 ± 0.015 | E2 0.391 ± 0.017 | E3 0.413 ± 0.019



Note: SE = SD of 20 optimizer-step means / $\sqrt{20}$; each mean averages 8 completions.

Descriptive only: no adjustment for serial, overlapping-window, or between-run dependence; not a confidence interval.